

Where to Look Next: A Data-Landscape & Opportunity Scoping Report for Strong-Lens Discovery

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Internal Scoping Report

Abstract

The agentic-lensing program has searched the DESI Legacy optical-imaging footprints (DR7–DR10, $\sim 19,000 \text{ deg}^2$) and the DESI DR1 spectroscopic catalogue (28 M fibres) about as hard as current methods allow, yielding $\sim 6,500$ lens candidates across CNN imaging finders, spectroscopic friends-of-friends search, and difference-imaging time-domain search. This report asks the complementary question: *which public telescope datasets have **not** been exhaustively searched for strong lenses, or have no published lens results?* We scanned the full multi-wavelength survey landscape with a deterministic multi-agent research workflow (8 survey-family researchers, adversarial red-team verification of every flagged opportunity, priority-scoring synthesis, completeness critic; 51 agents, $\sim 2.2 \text{ M}$ tokens). Of 41 challenged opportunities, 34 were confirmed, 6 downgraded, and 1 refuted. The headline finding is that the easy same-modality, in-footprint opportunities are gone — the major peer surveys (HSC PDR3, KiDS DR5, DES Y6, UNIONS, Euclid Q1) are already searched and published — and the genuine opportunities are: (i) *virgin off-modality reservoirs* (blind radio and submm/mm lens searches have *never* been done at catalogue scale: VLASS, the ALMA archive, ALMACAL, the faint-DSFG tail of Herschel/SPT-3G/ACT); (ii) *drop-in re-points* of the existing pipelines — but *all four* first-scan leads here (HST, VHS, GAMA, DELVE DR2) were downgraded or refuted on follow-up dives (§1, §7), leaving only prototype-scale survivors (time-domain ATLAS/ZTF, DELVE DR3-depth, JWST COSMOS-Web, J-PLUS/S-PLUS); (iii) a *foundation-model re-mine* of the fibre-collision-blocked SDSS/BOSS spectroscopy; and (iv) *pre-positioning* for the megasurveys (Euclid DR1 Wide, Rubin/LSST, Roman, DESI DR2). The lens-finding itself is out of scope; this report only maps where the un-mined data is, how to access it, and how each opportunity fits the program’s existing methods.

Provenance. Every GREEN/YELLOW rating below was challenged by an independent skeptic agent on four axes (*already searched? public now? new sky? method fits?*); a GREEN survives only with ≥ 2 corroborating sources. Numbers are workflow estimates; the companion Markdown report (OPPORTUNITIES.md) carries the same content with inline source links, and the workflow script is archived at `workflow.js`.

1 Executive summary

The program’s searched footprints are the *baseline* (§2), not opportunities. The real opportunities fall into four buckets:

Table 1: Top opportunities (priority score / tier / pipeline fit / access effort 1–5).

	Score	Opportunity	Tier	Fit	One-line
	78	VLA (VLA Sky Survey)	new-tooling	New	33,885 deg ² radio; no blind search ever;
	78/75	ALMA archive + ALMACAL	new-tooling	New	Petabyte of mm/submm interferometry
75/62→	55/45 [‡]	ATLAS/ZTF time-domain	drop-in	Diff	[‡] <i>downgraded, strongest survivor</i> (§1): r
	~55 [‡]	HST general archive (HLA)	drop-in	CNN	[‡] <i>downgraded on deep-dive</i> (§1): comple
	64→	low [‡]	<i>refuted</i>	FoF	[‡] <i>REFUTED</i> (§1): two-fibre FoF can't r
	63→	low [‡]	<i>refuted</i>	CNN(NIR)	[‡] <i>REFUTED</i> (§1): too shallow to see ar
	66/63	SPT-3G / Herschel faint tail	new-tooling	New	Large undelivered lensed-DSFG popula
62 / 61→	45 [‡]	JWST COSMOS-Web; DELVE DR2/3	drop-in	CNN	JWST: deeper space. [‡] DELVE DR2 <i>do</i>
	70–58	Euclid DR1 / Rubin / Roman / DESI-DR2	future-watch	—	The megasurvey prizes; pre-position wi

1. **Virgin off-modality reservoirs (highest novelty; needs new tooling).** Radio and submm/mm sky has never had a blind catalogue-scale strong-lens search — only forecasts (McCarty and Connor, 2024; McKean et al., 2008) and known-lens cross-matches (Martinez et al., 2024) exist. **VLA** (33,885 deg², public), the **ALMA Science Archive+ALMACAL** (Oteo et al., 2016), and the faint-DSFG tail of **Herschel/SPT-3G/ACT** are genuine green fields. VLA uniquely can also *radio-confirm the program’s existing ~3,900 optical candidates*.
2. **Drop-in re-points (public now).** *This bucket did not survive scrutiny.* All four first-scan leads (**HST archive**, **VHS**, **GAMA DR4**, **DELVE DR2**) were downgraded or refuted on dedicated deep-dives (§1). The honest survivors are narrow and prototype-scale: time-domain diff-imaging on **ATLAS/ZTF** (the most robust), the **DELVE DR3** depth gain + low- $|b|$ frontier, **JWST COSMOS-Web**, and **J-PLUS/S-PLUS**. There is no large fresh-public-data harvest with current methods (§7).
3. **Foundation-model re-mine of legacy spectroscopy.** ~5 M public **SDSS/BOSS/eBOSS** spectra cannot be pair-searched (the 55"/62" fibre-collision radius forbids two fibres on a galaxy-scale pair) and the single-fibre emission-residual channel is exhausted (Bolton et al., 2006; Brownstein et al., 2012; Shu et al., 2017; Talbot et al., 2021) — but a SpectrumFM-style re-mine of the full spectral set is an open, build-once lever.
4. **Pre-position for the megasurveys.** **Euclid DR1 Wide** (public 2026-10-21), **Rubin/LSST** (DP2 ~2026 Q3; world-public alerts), **Roman HLWAS** (launch by 2027; ~160,000-lens forecast, Holloway et al. 2025), and **DESI DR2** spectra (~2027) are the eventual prizes.

Figure 1 ranks the opportunities; Figure 2 shows where the virgin sky is. Table 1 lists the highest-priority opportunities.

Pressure-test log (deeper follow-up dives)

Individual opportunities are being stress-tested with dedicated 4-agent deep-dives (inventory + prior-coverage + pipeline → adversarial red-team); these *supersede* the first-scan ratings where they disagree. **HST general archive: DOWNGRADED** (first-scan score 68 → ~55). The dedicated dive (companion HST_ARCHIVE_DEEPPDIVE.md, 6 corroborating sources) found the “hundreds, never ML-swept” premise substantially false: HST is pencil-beam (~2.5–3 deg² contiguous deep mosaic; whole-archive secure population only ~250–350), and the archive is *not virgin* — Hubble Asteroid Hunter II (Garvin et al., 2022) already crowd-scanned ~27 deg² (198 new) and COSMOS was already CNN-searched (Pourrahmani et al., 2018) and JWST-mined (COWLS). Realistic **net-new secure yield: tens~150** via a completeness re-mine, not virgin discovery; best framing is a cheap HST→Euclid/Roman space-PSF CNN readiness prototype.

VHS: REFUTED (first-scan score 63; 8 corroborating sources; companion `VHS_DEEPDIVE.md`). Both load-bearing claims fail. (i) *Depth kill-shot*: VHS ($J_{AB} \sim 20.8$, $K_{s,AB} \sim 20.0$; deepest $J \sim 21.4$) is ~ 3 – 4 mag too shallow and has no blue bands, so faint blue arcs fall below the limit — a VHS CNN sees the deflector, not the lens (no blind wide-NIR galaxy–galaxy catalogue has *ever* been published; all wide-NIR lensing is submm-counterpart or optical-colour use). (ii) *Not fresh*: DECam reaches $\delta = -90^\circ$, and DELVE DR2 ($\sim 21,000 \text{ deg}^2$ griz) + DES + Legacy DR10 already blanket the southern sky 3–4 mag deeper and mostly lens-searched, so net NIR-only fresh usable sky is ~ 0 – 10% (the unusable plane). Realistic net-new yield ~ 0 –low-tens. The southern residual it points to is re-running the optical CNN on **DELVE DR2** griz, not a NIR search. **GAMA DR4: REFUTED** (first-scan score 64; 6 sources; `GAMA_DELVE_DEEPDIVE.md`). GAMA’s multi-pass tiling does beat the fibre-collision wall, but that is a non-sequitur for lensing: galaxy–galaxy lenses have $\theta_E \sim 0.5$ – $2''$, so the source lands inside one $2''$ fibre — the two-fibre FoF channel cannot reach them (only wide/group pairs). The channel that works is the single-fibre blended spectrum, and Holwerda et al. (2015) already did it (104 candidates); the same channel on DESI just yielded 4,110 (arXiv:2512.04275). Net-new galaxy–galaxy ≈ 0 .

DELVE DR2: DOWNGRADED (first-scan score 61; 5 sources). The “fresh sky” premise fails: DELVE DR2 tops out at $\delta \sim +30$ (below Legacy DR10’s $+32$ ceiling), and Legacy DR10 (Inchausti Reyes et al., 2025) already CNN-searched $\sim 14,000 \text{ deg}^2$ south of $+32$ and ingested the same DELVE/DeROSITAS exposures, so ~ 80 – 88% is the same photons already searched. Genuinely fresh-and-deep residual is only $\sim 2,000$ – $4,000 \text{ deg}^2$ of low- $|b|$ sky; the real levers are the **DELVE DR3** deeper coadd (~ 0.5 – 0.7 mag) and the low- $|b|$ frontier (tens–few-hundred low-grade, eclipsed by Rubin within ~ 2 – 3 yr).

ATLAS/ZTF time-domain: DOWNGRADED — the strongest survivor (first-scan score 75/62; 16 sources; `ATLAS_ZTF_DEEPDIVE.md`). The thesis fails on three of four axes but keeps a real niche and a forward-looking role. ZTF’s gISN alert content is *saturated* — a full-archive search (Magee 2023) returned 0, an AMPEL pipeline runs live, the group’s *own* Park/Huang 2026 finder is validated on ZTF, and 4 brokers classify every alert; both real ZTF gISNe were caught live by the magnification method, not at lens positions, so position-targeted diff-imaging adds $\ll 1$. Lensed-QSO discovery is blocked by the $< 2''$ resolution wall (~ 90 – 95% unresolved at ZTF’s $\sim 2''$ PSF); net-new ≈ 0 . “Diff-imaging drops straight in” is *refuted* — ZTF/ATLAS already emit difference-image alerts + forced photometry. ATLAS is the one genuine under-searched gISN channel but is depth-capped to a handful of extreme events. *What survives, and uniquely, is a constructive reframing: a known-lens watchlist + a SpectrumFM spectral gate that doubles as a Rubin-era dress-rehearsal* (§7).

Meta-finding (read first): *all five* current-data “fresh public data, existing method” leads led with in the first scan (HST, VHS, GAMA, DELVE DR2, ATLAS/ZTF) have now been downgraded or refuted (2 refuted, 3 downgraded). The southern DECam optical sky, the DESI/SDSS spectroscopy, and the ZTF time-domain stream are far more thoroughly mined — by the group itself (Legacy DR10, Hsu 2025, Park/Huang 2026) and the community (HAH-II, Holwerda 2015, DESI single-fibre, KiDS/HSC CNNs, Magee 2023, the Goobar group, 4 brokers) — than a first-pass scan credits, and each alternative modality hits a hard wall. **There is no large fresh-public-data harvest reachable with current techniques**; the one lead that points somewhere constructive is the ATLAS/ZTF→Rubin precursor. See the synthesis in §7.

2 Scope & baseline (what is excluded)

This report *excludes* the footprints the program has already mined (indexed in `reproductions/REPRODUCTIONS.md`); these are the baseline, not opportunities:

- **DESI Legacy Imaging DR7/8/9/10** (DECaLS+BASS+MzLS, $grz/+i$): optical-CNN finders (Huang et al., 2020, 2021; Inchausti Reyes et al., 2025; Storfer et al., 2024) $\rightarrow \sim 3,900$ candidates.
- **DESI DR1 spectroscopy** (28 M redshifts): spectroscopic FoF/pair (Hsu et al., 2025) + quasar autocorrelation (Dawes et al., 2023) $\rightarrow \sim 2,600$ candidates.
- **DECam multi-epoch** over the lens footprint: difference imaging (Sheu et al., 2023, 2024).
- **HST/MAST, Keck NIRES, VLT/MUSE, Euclid Q1**: follow-up/confirmation only.

The three in-house methods (used for the *fit* column): **optical-cnn** (CNN on $grz(i)$ cutouts; retrained for NIR/space PSF as a config change), **spectro-fof** (redshift-pair/emission-residual search), **diff-imaging** (multi-epoch). Anything else (radio source-matching, uv-plane lens modelling, submm DSFG selection, IFU, astrometric multiplets, foundation-model re-mine) is **new-tooling**.

Rating rubric. RAG (verified): **G** largely unsearched & public; **Y** partial / searched-but-unpublished / future-but-high-value; **R** exhausted or unviable. **Priority** = $0.75 \cdot \text{opportunity} + 0.25 \cdot \text{actionability}$ (opportunity weighs verified RAG, fresh sky, yield; actionability weighs pipeline fit, access effort, public availability — so scientific opportunity dominates convenience). **Tier:** drop-in-now / new-tooling-now / future-watch / low.

3 Master opportunity table

All 55 datasets, ranked by priority (Table 2). Avail: P=public, E=embargoed, F=future. Fit: CNN=optical-cnn, FoF=spectro-fof, Diff=diff-imaging, New=new-tooling.

Table 2: Master opportunity table (verified, priority-ranked).

Pri	RAG	Fit	Av	Dataset	Note
78	G	New	P	VCLASS	33,885 deg ² radio; no blind lens search ever
78	G	New	P	ALMA Science Archive	PB interferometry, never systematically lens-searched
75	G	New	P	ALMACAL	$\sim 1,000$ calibrator sightlines, virgin (no proprietary period)
75 $\rightarrow$ 55 [‡]	Y	Diff	P	ATLAS (Tonry) ‡	[‡] <i>downgraded</i> (§1): the one genuine g1SN gap, but depth-capped to a handful of extreme-mag events
70	Y	CNN	F	Roman HLWAS+HLTDS	$\sim 160,000$ -lens forecast (launch by 2027)
69	Y	Diff	E	Rubin / LSST	$\sim 10^5$ full survey; world-public alert stream
68	Y	CNN	F	Euclid DR1 Wide	$\sim 7,000$ A/B forecast; public 2026-10-21
$\sim 55^\dagger$	Y	CNN	P	HST general archive (HLA)	[†] <i>downgraded</i> (§1): not virgin (HAH-II scanned ~ 27 deg ²); tens-150 net-new via re-mine
68	Y	FoF	F	4MOST (south)	only large fresh <i>southern</i> spectroscopy
66	Y	New	P	Herschel-ATLAS	faint sub-threshold DSFG residual $O(10^3)$
65	Y	New	P	HerMES+HeLMS+HerS	faint-DSFG residual

Table 2 – continued

Pri	RAG	Fit	Av	Dataset	Note
64→low [‡]	R	FoF	P	GAMA DR4 [‡]	[‡] <i>REFUTED</i> (§1): FoF can't reach G-G θ_E ; blended already done (Holwerda 2015); ~ 0 net-new G-G
63→low [‡]	R	CNN	P	VHS [‡]	[‡] <i>REFUTED</i> (§1): $\sim 3\text{--}4$ mag too shallow for arcs (deflector-only); DECam covers the sky deeper
63	Y	New	P	SPT-3G emissive catalog	deeper mm; 4,303 candidates already extracted
62	Y	CNN	P	JWST COSMOS-Web	re-searchable deeper than the COWLS visual pass
62→45 [‡]	Y	Diff	P	ZTF [‡]	[‡] <i>downgraded</i> (§1): gLSN field saturated (Magee→0; Park/Huang 2026; 4 brokers); lensed-QSO blocked by $< 2''$ wall; net-new ≈ 0
61→45 [‡]	Y	CNN	P	DELVE DR2/DR3 [‡]	[‡] <i>DOWNGRADED</i> (§1): $\sim 80\text{--}88\%$ already in Legacy DR10; DR3-depth + low- $ b $ frontier are the real levers
60	Y	CNN	E	UNIONS	z-band/footprint/CFIS-u residual (GLUE I ran deep gri)
60	Y	CNN	P	J-PLUS / S-PLUS	novel 12-band narrow-band selection
60	Y	FoF	E	WEAVE	fresh northern spectroscopy (annual DRs)
60	Y	FoF	E	Subaru PFS	$\sim 100\%$ fresh as spectroscopy
59	Y	FoF	F	DESI-II / Spec-S5	deeper/higher- z over DESI footprint (future)
58	Y	FoF	E	DESI DR2 spectra	$\sim 50\%$ new objects; ~ 2027
58	Y	New	P	ACT DR6 mm catalog	faint lensed-DSFG residual over 19,000 deg ²
57	Y	FoF	P	LAMOST DR11/12	$\sim 30\%$ fresh objects vs SDSS NGC
57	Y	Diff	P	DECam multi-epoch (new)	DES-SN + DELVE-DEEP + archive; fresh epochs
56	Y	CNN	P	VISTA VIKING	NIR modality gap (=KiDS sky)
56	Y	New	P	ASKAP EMU + RACS	$\sim 20,000$ deg ² fresh southern radio
54	Y	CNN	F	HSC-SSP PDR4	increment over PDR3 (release date unknown)
53	Y	CNN	P	UKIDSS LAS	modest NIR residual
53	Y	New	P	Planck PHz/PCCS2	all-sky submm residual
52	Y	CNN	P	VIDEO	deep NIR, tiny area
51	Y	CNN	P	VST-ATLAS DR4	u-band differentiator; $\sim 0\%$ fresh sky
50	Y	New	P	LOFAR LoTSS DR2	sub-arcsec ILT uv-plane = high ceiling (unbuilt)
50	Y	New	P	ALMA Lensing Cluster Survey	cluster-field mm; low galaxy-scale content
48	Y	New	P	SDSS/BOSS/eBOSS (+MaNGA)	foundation-model re-mine only (fibre-collision-blocked)
48	Y	New	P	MeerKAT MIGHTEE/MALS	small/medium-area radio
47	Y	New	P	Gaia GraL/GravLens	astrometric lensed-QSO; largely worked
46	R	CNN	P	SkyMapper DR4	<i>refuted</i> : $\sim 1.5''$ seeing / shallow $\Rightarrow$ unviable
44	R	CNN	P	KiDS DR5	searched; incremental only
44	Y	New	P	Gaia Photometric Alerts	bright limit dominates
42	R	CNN	P	JWST other deep fields	now swept (AnomalyMatch / COWLS / Nagam 2025)
41	R	CNN	P	Pan-STARRS1 3π DR2	shallow, heavily overlapped
36	R	CNN	P	HSC-SSP PDR3	searched end-to-end (SuGOHI/HOLISMOKES)

Table 2 – continued

Pri	RAG	Fit	Av	Dataset	Note
35	R	CNN	P	DECaLS-south DR10 <i>i</i> -band	program baseline
35	R	FoF	P	2dFGRS + 6dFGS	shallow/low- <i>z</i>
34	R	CNN	P	HST COSMOS (ACS)	searched
34	R	FoF	P	DEVILS DR1	tiny area
33	R	CNN	P	Euclid Q1	exhausted benchmark
32	R	CNN	P	DES Y6 / DR2	exhausted (Space Warps / Jacobs / Rojas)
32	R	CNN	P	Southern Pan-STARRS strip	~0% fresh
32	R	CNN	P	UltraVISTA	over-studied COSMOS
30	R	New	P	SPT-SZ DSFG sample	bright selection exhausted
29	R	New	P	FIRST + NVSS (CLASS/JVAS)	legacy; low residual
23	R	New	P	VVV / VVVX	Zone of Avoidance

4 Findings by family

4.1 Wide optical ground imaging — mostly RED

The southern DECam optical sky (DES, DESI-Legacy DR10-south, DELVE DR1) is the most multiply-searched region in this study. DES Y6 is exhausted (Jacobs et al., 2019a,b; O’Donnell et al., 2025; Rojas et al., 2022); HSC PDR3 is combed end-to-end (Cañameras et al., 2024; Schuld et al., 2025; Sonnenfeld et al., 2018); KiDS DR5 is done (Grespan et al., 2024; Li et al., 2020; Petrillo et al., 2019); UNIONS deep *gri* was run by GLUE I (Savary et al., 2022; UNIONS Collaboration, 2025). **DELVE DR2** (first-scan score 61) was **downgraded** on deep-dive (§1): it tops out at $\delta \sim +30$ and Legacy DR10 (Inchausti Reyes et al., 2025) already searched $\sim 14,000 \text{ deg}^2$ south of $+32$ using the same ingested DELVE exposures, so $\sim 80\text{--}88\%$ is already-searched sky; the genuine levers are the **DELVE DR3** deeper coadd and the low- $|b|$ frontier (a niche, not a fresh harvest; GAMA_DELVE_DEEPDIVE.md). SkyMapper DR4 was **refuted** (seeing/depth make galaxy-scale lensing unviable).

4.2 Wide-area NIR ground imaging — a modality gap with a *physical* cause

Nobody has run a blind wide-NIR lens search, and the VHS deep-dive shows *why*: NIR is too shallow to see the arcs. **VHS (R REFUTED)** on deep-dive, 8 sources; first-scan score 63 — I initially rated it the top “fresh far-south” drop-in). (i) *Depth kill-shot*: $J_{AB} \sim 20.8/K_{s,AB} \sim 20.0$ (deepest $J \sim 21.4$) is $\sim 3\text{--}4$ mag too shallow, with no blue bands, so faint blue arcs fall below the limit — a VHS CNN sees the deflector, not the lens (no blind wide-NIR galaxy–galaxy catalogue has ever been published). (ii) *Not fresh*: DECam reaches $\delta = -90^\circ$; DELVE DR2 ($\sim 21,000 \text{ deg}^2$ griz) + DES + Legacy DR10 blanket the southern sky $3\text{--}4$ mag deeper and mostly lens-searched, so net usable fresh sky is $\sim 0\text{--}10\%$ (the unusable plane). Realistic yield $\sim 0\text{--}low\text{-tens}$; the southern residual it points to is re-running the optical CNN on **DELVE DR2** griz, not a NIR search. **VIKING** is deeper ZYJHKs but its footprint *is* the already-searched KiDS sky; **VIKING/UKIDSS/ VIDEO** inherit the same NIR depth limit (VHS_DEEPDIVE.md).

4.3 Space high-resolution imaging — mixed

The contiguous deep fields are now swept (Nagam et al. 2025; Nersesian et al. 2025; AnomalyMatch 2026). The **HST general archive** looked like the best space drop-in, but the dedicated pressure-test (§1) **downgraded** it: HST is pencil-beam and not virgin (Hubble Asteroid Hunter II (Garvin et al., 2022) already scanned $\sim 27 \text{ deg}^2$; COSMOS was already CNN-searched (Faure et al., 2008; Pourrahmani et al., 2018) and JWST-mined), so the realistic net-new yield is tens- ~ 150 secure via a completeness re-mine, not hundreds. **JWST COSMOS-Web** is re-searchable deeper than COWLS. The astrometric Gaia lensed-QSO search is mature (Krone-Martins et al., 2025; Lemon et al., 2019). Euclid Q1 is exhausted (Euclid Collaboration, 2026; Euclid Collaboration et al., 2025); Euclid DR1 and Roman are future-watch.

4.4 Optical spectroscopy — FoF is the wrong tool; the blended channel is, and it is mined

GAMA DR4 (first-scan score 64) was **refuted** on deep-dive (§1). Its multi-pass tiling does beat the fibre-collision wall, but galaxy-galaxy lenses have $\theta_E \sim 0.5\text{--}2''$, so the source lands inside one fibre — the two-fibre FoF channel cannot reach them. The channel that works is the single-fibre blended spectrum, and Holwerda et al. (2015) already executed it on GAMA; the same channel on DESI just yielded 4,110 candidates (DESI Collaboration, 2025). So GAMA’s residual is wide/group-pairs + a confirmation/testbed role, and the scalable spectroscopic lever is a foundation-model blended-spectrum re-mine, not FoF pairs. **SDSS/BOSS/eBOSS** (~ 5 M public spectra, trivial access) is otherwise blocked: the pairwise method is fibre-collision-limited and the single-fibre residual channel is exhausted (Bolton et al., 2006; Brownstein et al., 2012; DESI Collaboration, 2025; Shu et al., 2017; Talbot et al., 2021) — the only open lever is a foundation-model re-mine (the natural SpectrumFM science target). **LAMOST** adds $\sim 30\%$ fresh objects over SDSS NGC at low resolution.

4.5 Radio continuum — the headline new-tooling opportunity (GREEN)

No blind catalogue-scale radio strong-lens search has ever been published; the literature is forecasts (McCarty and Connor, 2024; McKean et al., 2008; Rezaei et al., 2022), known-lens cross-matches (Martinez et al., 2024), or legacy lobe surveys (Browne et al., 2003; Phillips et al., 2001). **VLA** ($33,885 \text{ deg}^2$, public) offers a radio-source ML cross-match for blind discovery *and* a way to radio-confirm the program’s existing optical candidates. The **LOFAR LoTSS** sub-arcsec ILT (uv-plane) path is the unbuilt high-yield lever [$O(10^2\text{--}10^3)$]; **ASKAP EMU/RACS** adds fresh southern radio sky.

4.6 Submm / mm — virgin modality plus a faint residual

Bright-source submm lensing is famously complete (RED: Cañameras et al. 2015; Everett et al. 2020; Nayyeri et al. 2016; Negrello et al. 2010, 2017; Reuter et al. 2020; Vieira et al. 2013; Wardlow et al. 2013), but the **ALMA Science Archive+ALMACAL** (Oteo et al., 2016) hold a petabyte of interferometry *never* systematically lens-searched (GREEN; uv-plane lens-finding). **SPT-3G** already has 4,303 extracted candidates (Archipley et al., 2024); **ACT DR6** (Gralla et al., 2020) and **Planck** (Trombetti et al., 2021) have similar faint residuals.

4.7 Time-domain — DOWNGRADED, but the strongest survivor (reframe to a Rubin precursor)

ATLAS/ZTF were **downgraded** on deep-dive (§1; ATLAS_ZTF_DEEPDIVE.md, 16 sources). ZTF’s glSN content is *saturated* (Magee 2023 full-archive →0; AMPEL live; the group’s own Park/Huang 2026 finder; 4 brokers (Goobar et al., 2023; Townsend et al., 2025)), and “diff-imaging drops straight in” is *refuted* — ZTF/ATLAS already emit difference-image alerts + forced photometry. Lensed-QSO discovery is blocked by the $< 2''$ resolution wall. **ATLAS** is the one genuine under-searched glSN channel but is depth-capped to a handful of extreme events. The defensible, forward-looking play is a known-lens **watchlist** on the live broker/forced-phot streams + a **SpectrumFM spectral gate** — a dress-rehearsal for **Rubin/LSST** (the eventual prize, $\sim 100\times$ rate, world-public alert stream, $\sim 1\text{--}2$ yr away).

Red-team outcomes. Of 41 challenged opportunities: 34 CONFIRMED, 6 DOWNGRADED (JWST-deep-fields→red, SPT-3G→yellow, Gaia-GraL→yellow, ATLAS→yellow, HSC-PDR4→yellow, VHS→yellow), 1 REFUTED (SkyMapper→red). The three surviving GREENs (VLASS, ALMA archive, ALMACAL) each carry ≥ 5 corroborating sources.

5 Cross-cutting analysis

Footprint overlap. The most multiply-searched region is the southern DECam optical sky ($\text{DES} \cup \text{DESI-Legacy-DR10-south}$, which ingested DELVE/DeROSITAS exposures, $\cup$ DELVE DR1). VST-ATLAS, SkyMapper and the southern PS1 strip all sit *inside* this footprint ($\sim 0\%$ new sky). COSMOS ($\sim 2 \text{ deg}^2$) is the single most-searched patch. The genuinely virgin reservoirs are: fresh southern radio (EMU/RACS, ACT-DR6); deeper mm (SPT-3G); the submm interferometric modality (ALMA/ALMACAL, positionally decoupled from optical); all-sky time-domain *content*; and the future megasurveys. Key dedup facts: VIKING $\approx$ KiDS sky; UKIDSS LAS $\subset$ SDSS/DECaLS-north; DELVE DR2/3 $\approx$ DES $\cup$ Legacy-DR10 (only $\sim 10\text{--}20\%$ net-new).

Method $\times$ modality. Drop-in now (post-pressure-test): optical-cnn → DELVE DR3 (depth) + low- $|b|$, J-PLUS/S-PLUS, JWST COSMOS-Web; spectro-fof → LAMOST + GAMA wide-pairs only; diff-imaging → ATLAS, ZTF, DECam-multi-epoch. *Four first-scan candidates failed scrutiny (§1): HST archive (downgraded), VHS/wide-NIR (refuted — too shallow for arcs), GAMA DR4 (refuted for G-G — FoF can’t reach θ_E ; blended already done), and DELVE DR2 as a fresh harvest (downgraded — already in Legacy DR10).* Needs a build: SDSS/BOSS (foundation-model blended-spectrum re-mine); all radio (source-match or uv-plane); all submm (faint-DSFG selector or uv-plane); Gaia (astrometric ML). The highest-value builds are the VLASS cross-match, the ALMA/ALMACAL uv-plane lens-finder, the SPT-3G/Herschel faint-DSFG selector, and the SpectrumFM SDSS/BOSS re-mine.

Searched-but-unpublished (direct-inquiry targets). (1) SPT-3G’s 4,303 strong-SMG candidates (Archipley et al., 2024) were never published as a lens catalogue; (2) the Euclid Discovery Engine consortium will sweep DR1 Wide internally around the Oct-2026 release; (3) the GLUE/UNIONS group is searching embargoed multi-band data; (4) SuGOHI/HOLISMOKES will run finders on the HSC PDR4 increment; (5) ACT DR6 has no published lensed-DSFG search beyond the brightest-30; (6) Planck PCCS2 faint candidates ($\sim 50\%$ unconfirmed); (7) Gaia GravLens FPR small-separation candidates lack a public confirmed catalogue.

6 Gaps the completeness critic found

The critic flagged ~ 16 genuinely-missed datasets in three *new families* the original 8 missed — recommended as the next scan’s seed list (Table 3).

Table 3: Critic-flagged additions (next-scan seed list).

Added dataset	New family	Why it matters
WISE/unWISE/CatWISE2020 (Marocco et al., 2021)	mid-IR all-sky cross-match	universal lens-vett
eROSITA eRASS1 DR1 (Merloni et al., 2024)	X-ray	12,247-cluster cata
NOIRLab Source Catalog DR2 (Nidever et al., 2021)	ground optical cross-match	unified all-sky DE
JCMT S2CLS (Geach et al., 2017)	deep ground submm	high-res lensed-DS
HST/JWST grism; Euclid NISP grism	space slitless spectroscopy	blind emission-line
GMRT TGSS / GLEAM / Apertif	low-freq / southern radio	steep-spectrum len
CFHTLS / CFHTLenS / SL2S	deep ground optical + ML training	the canonical Ring
VLT/MUSE (+KMOS) ESO archive	optical IFU spectroscopy	de-facto confirmati
Quaia / Gaia×WISE (Storey-Fisher et al., 2024)	all-sky lensed-QSO cross-match	ready-made quasar
Spitzer SWIRE/SERVS; J-PAS; Stripe 82; Chandra/4XMM; SO/CMB-S4	various	secondary addition

7 Ranked action shortlist

Tier A — drop-in now (public data, existing method), *post-pressure-test*. After five deep-dives this tier is *much smaller than the first scan implied* and is best treated as prototype-scale (§7). Ordered by what survived: (1) **ATLAS/ZTF reframed as a Rubin precursor** (the strongest survivor, but *downgraded*): do not re-run diff-imaging — build a known-lens **watchlist** on the live broker/forced-phot streams + a **SpectrumFM spectral gate** + an ATLAS extreme-mag glSN sweep; best-justified because it is forward-looking (scales onto Rubin); (2) **DELVE DR3** deeper pass + low- $|b|$ frontier (*downgraded* from DR2 “fresh harvest”); (3) **JWST COSMOS-Web** → deeper CNN (small field); (4) **J-PLUS/S-PLUS** narrow-band (speculative); (5) **HST general archive** (*downgraded*: completeness re-mine, best as the HST→Euclid/Roman readiness prototype); (6) **GAMA** (wide-pair + confirmation layer + blended testbed) and **LAMOST**, DECam-multi-epoch — supporting. *Refuted/downgraded out of the lead: VHS + wide-NIR (too shallow for arcs), GAMA DR4 for galaxy-galaxy (FoF can’t reach θ_E), DELVE DR2 as a fresh harvest (already in Legacy DR10), and ZTF for blind glSN/lensed-QSO discovery (saturated + resolution wall).*

Synthesis after five pressure-tests (the honest bottom line). *Every* current-data “fresh public data + existing method” lead from the first scan was downgraded or refuted (HST, VHS, GAMA, DELVE DR2, ATLAS/ZTF — 2 refuted, 3 downgraded). (1) **Small current-data program (prototype-scale, now):** the **ATLAS/ZTF**→Rubin precursor (watchlist + SpectrumFM gate + ATLAS extreme-mag glSN sweep — the best-justified item because it is forward-looking); **DELVE DR3** depth + low- $|b|$; **GAMA** confirmation/testbed; **JWST COSMOS-Web**; **J-PLUS/S-PLUS** — realistic combined net-new of order low hundreds, much of it transients. (2) **The real, large opportunities** (unchanged — never “existing-technique” claims): the new-tooling green fields (**VLASS**, the **ALMA archive/ALMACAL**, and the **SDSS/BOSS SpectrumFM** blended-spectrum re-mine) and pre-positioning for **Euclid DR1 / Rubin / Roman / DESI DR2**, where the 10^3 – 10^5 populations live. (3) **Posture:** run the current-data items as readiness prototypes for the space-PSF CNN and the alert-stream diff-imaging, while investing the real effort in the new-tooling builds and megasurvey pre-positioning.

Tier B — new-tooling now (public, real opportunity, build required). (1) **VCLASS** radio cross-match (+ radio-confirm existing optical lenses); (2) **ALMA archive+ALMACAL** uv-plane lens-finder; (3) **SPT-3G+Herschel** faint-DSFG selector (start by requesting the Archipley 4,303-candidate list); (4) **SDSS/BOSS** foundation-model re-mine (the SpectrumFM science target); (5) **EMU/RACS, LoTSS-ILT, ACT DR6, Planck.**

Tier C — pre-position for the future (inquiries + readiness now). **Euclid DR1 Wide** (2026-10-21; request early/joint consortium access); **Rubin/LSST** (connect to a broker; alert-stream diff-imaging is ready); **Roman HLWAS** (ready a space-PSF CNN; largest single ceiling); **DESI DR2** spectra (~ 2027 ; day-one FoF re-run); **4MOST** (the only large fresh *southern* spectroscopy).

8 Future watch-list & timeline

Table 4: Future releases with expected public-availability and recommended action.

Public	Dataset	Fit	Action
2026 Q3 (DP2)	Rubin/LSST (DR1 ~ 2027)	Diff/CNN	broker connection; alert-stream diff-imaging
mid-2026	4MOST	FoF	southern FoF plan
2026	WEAVE	FoF	monitor WAS public DRs
2026-10-21	Euclid DR1 Wide	CNN/grism	early/joint consortium inquiry
unknown	HSC-SSP PDR4	CNN	monitor; SuGOHI internal lists
~ 2027	Subaru PFS (SSP)	FoF	SMOKA after 18-mo proprietary
launch by 2027	Roman HLWAS+HLTDS	CNN/Diff	space-PSF CNN readiness
~ 2027	DESI DR2 spectra	FoF	day-one Hsu-2025 FoF re-run
2026–2031 ops	DESI-II / Spec-S5 (Schlegel et al., 2025)	FoF	long-horizon
online now	Simons Observatory LAT	New	monitor first mm catalogues

A Data-access cookbook (selected public-now opportunities)

Optical/NIR imaging. **DELVE DR2/3:** NOIRLab Astro Data Lab SIA https://datalab.noirlab.edu/sia/delve_dr3 + TAP <https://datalab.noirlab.edu/tap> (anon; account for bulk). **VHS:** Astro Data Lab <https://datalab.noirlab.edu/vhsdr5.php> (TAP/SIA, anon). **VIKING/UKIDSS/VIDEO:** WFAU VSA/WSA <http://horus.roe.ac.uk/vsa/>, <http://wsa.roe.ac.uk/>. **J-PLUS/S-PLUS:** <https://splus.cloud/> (+ splusdata API). **HSC PDR3:** <https://hsc-release.mtk.nao.ac.jp/doc/> (free registration). **VST-ATLAS:** ESO Phase 3 + WFAU OSA <http://osa.roe.ac.uk>.

Space imaging. **HST HLA:** <https://hla.stsci.edu/> (anon). **JWST COSMOS-Web:** <https://exchg.calet.org/cosmosweb-public/DR1/>; deep fields via DJA <https://dawn-cph.github.io/dja/>. **Euclid Q1/DR1:** ESA SAS <https://easidr.esac.esa.int/sas/> (free registration). **Gaia GraL:** Gaia Archive TAP <https://gea.esac.esa.int/archive/>.

Spectroscopy. **SDSS/BOSS/eBOSS:** https://www.sdss4.org/dr17/data_access/ (SAS bulk + CAS TAP). **GAMA DR4:** <http://www.gama-survey.org/dr4/> (TAP + bulk). **LAMOST:** <https://www.lamost.org/dr12/>. **DESI DR2:** <https://data.desi.lbl.gov/doc/releases/> (at release).

Radio. **VCLASS:** CIRADA/CADC cutout + catalogs <https://cirada.ca/vlasscatalogueq10>. **LoTSS DR2:** https://lofar-surveys.org/dr2_release.html. **EMU/RACS:** CSIRO CASDA <https://research.csiro.au/casda/>. **MALS:** <https://mals.iucaa.in>.

Submm/mm. **Herschel:** HeDaM <https://hedam.lam.fr/>. **SPT:** <https://pole.uchicago.edu/public/data/>. **ACT DR6:** <https://lambda.gsfc.nasa.gov/product/act/>. **Planck:** PLA <https://pla.esac.esa.int/>. **ALMA/ALMACAL/ALCS:** ASA <https://almascience.eso.org/asax/>, TAP <https://almascience.eso.org/tap> (+ ALminer); calibrators have no proprietary period.

Time-domain. **ATLAS:** forced-phot server <https://fallingstar-data.com/forcedphot/> (free reg). **ZTF:** IRSA <https://irsa.ipac.caltech.edu/Missions/ztf.html> + brokers. **Gaia Alerts:** <https://gsaweb.ast.cam.ac.uk/alerts/>. **Rubin:** <https://dp1.lsst.io/> (alerts world-public via brokers). **DECam multi-epoch:** NOIRLab <https://astroarchive.noirlab.edu/>.

Critic-added archives (next scan). WISE/unWISE/CatWISE2020 (IRSA); eROSITA eRASS1 <https://erosita.mpe.mpg.de/dr1/>; NSC DR2 (Astro Data Lab); JCMT S2CLS (CADC JSA); HST/JWST grism (MAST); VLT/MUSE (ESO/AMUSED); GMRT TGSS <https://tgssadr.strw.leidenuniv.nl/>; CFHTLS/SL2S (CADC); Quiaia (Zenodo/VizieR).

The deep-research workflow gathered 147 unique prior-search references; the subset that sets each survey's RED/YELLOW status is bibliographed below. See `OPPORTUNITIES.md` for the complete annotated index with source URLs.

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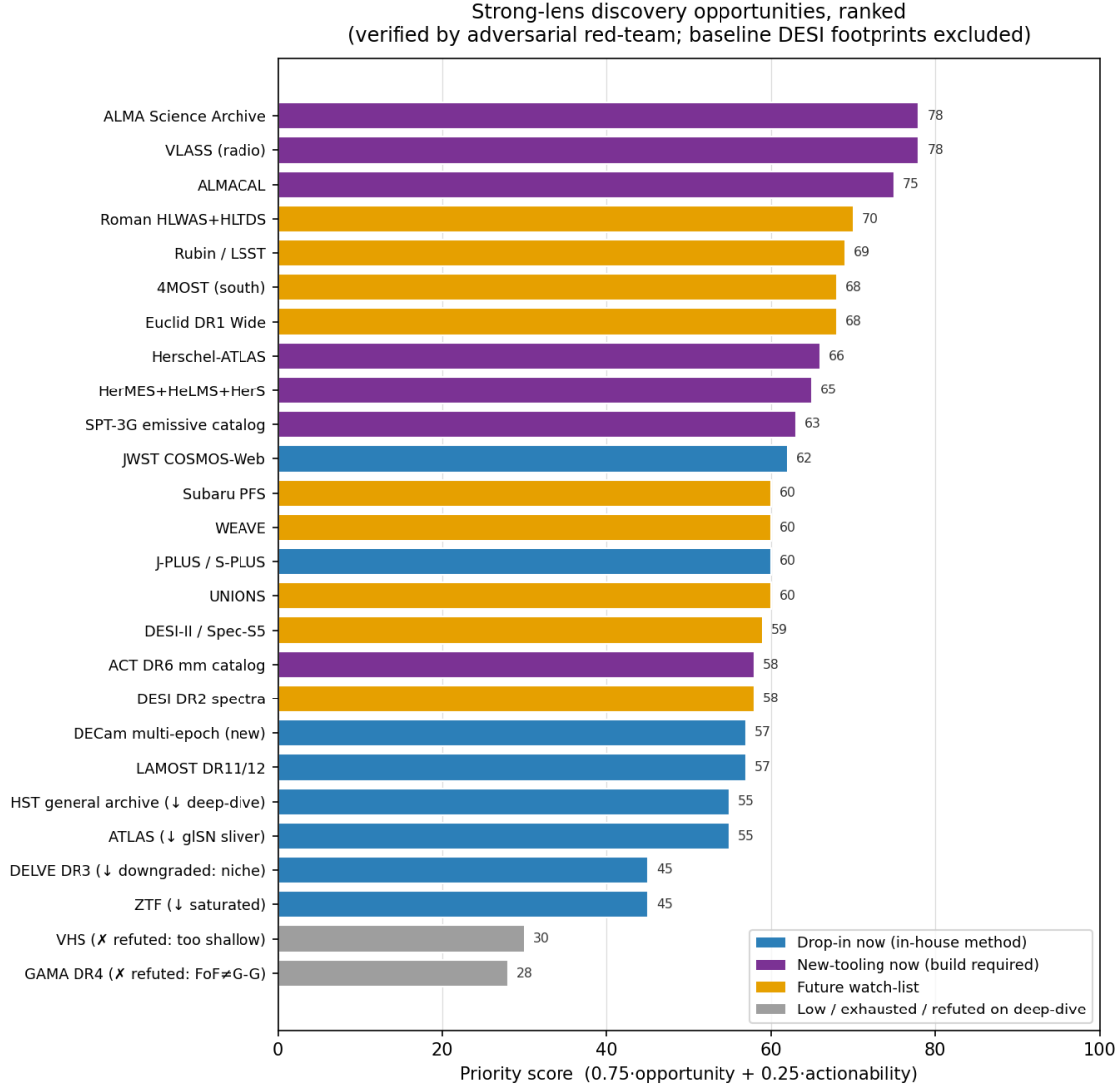


Figure 1: Verified opportunities ranked by priority score ($0.75 \cdot \text{opportunity} + 0.25 \cdot \text{actionability}$), coloured by tier. Baseline DESI footprints are excluded. Radio/submm reservoirs (purple) top the list on novelty despite needing new tooling; the blue drop-ins re-use existing pipelines.

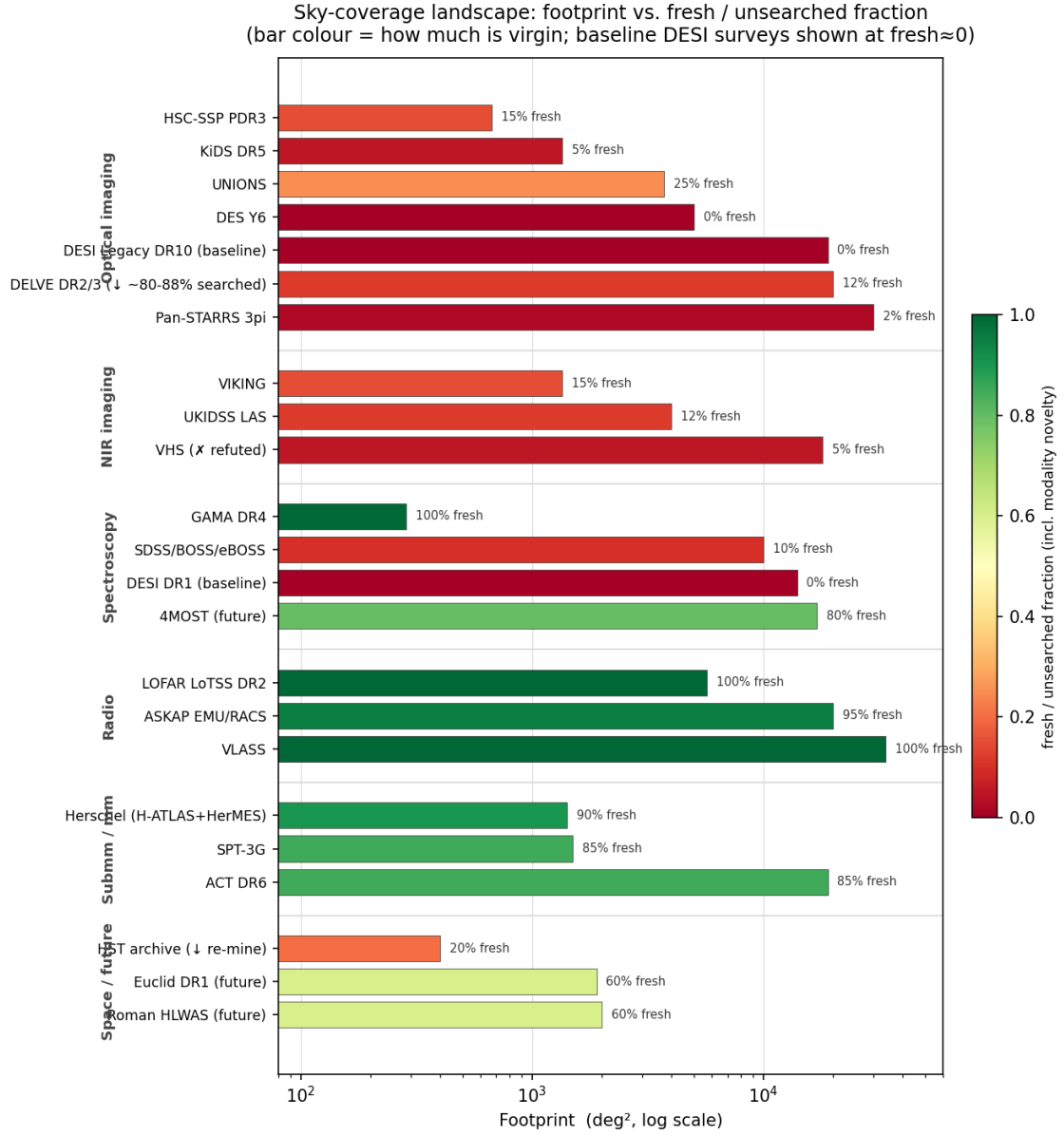


Figure 2: Sky-coverage landscape: footprint ($\log \text{deg}^2$) vs. fresh/unsearched fraction (bar colour; includes modality novelty), grouped by modality. Baseline DESI surveys sit at fresh $\approx$ 0 (red). The virgin reservoirs are wide-area radio (VLASS, EMU/RACS), the submm modality, and the future megasurveys.